

Sadia Humaira Salsabil

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RESEARCH INTEREST

- Quantum Systems Architecture
- Quantum Scattering
- Photonics

EDUCATION & STANDARDIZED TEST SCORES

North South University | Dhaka, Bangladesh

Oct 2018 - Sep 2022

B.Sc. in Electrical and Electronic Engineering

Minor in Mathematics

CGPA: 3.92/4.00 (Distinction: Summa Cum Laude)

Award: 50% Merit-Based Scholarship

IELTS: 7.5 (L: 9 | R: 7.5 | W: 6.5 | S: 7)

23 Aug 2024

GRE: 313 (Quant: 163 | Verbal: 150 | AWA: 3.5)

12 Dec 2024

PUBLICATION & PRESENTATION

Publication:

Salsabil, Sadia Humaira, Golam Dastagir Al-Quaderi, and M. R. C. Mahdy. "Quantum mechanical lateral force on an atom due to matter wave." Heliyon (2023): e23449. [Link](#)

Paper Presentation:

"Backstory of Matter Wave Manipulation & How to Conduct it without an Experimental Lab", Symposium, Quantum Tech. Fest, SIMATS School of Engineering, India, 23 February 2024

EMPLOYMENT

QWorld

Jun - Aug 2024

Designation: **Summer Project Intern**

- Description: Summer Quantum Project Intern under Mr. Shantanu Jha and Mr. Muhammad Waqar Amin. [Link](#)
- Responsibilities: Conducting original research on **Hamiltonian Engineering of Superconducting Qubit Circuits**, specifically gates on Kerr Cat qubits. Writing the research article.
- Skills: QuTiP
- Award: 2nd Best Project

North South University

Jan 2021 - Jan 2024

Designation: **Research Assistant**

Oct 2022 - Jan 2024 (16 months)

- Description: Research Assistant at *Optics Lab* ([Link](#)) of Dr. Mahdy Rahman Chowdhury in the Electrical and Computer Engineering Department.
- Responsibilities: Conducting original research on **Matter Wave Manipulation at an Atomic Scale**. Co-supervising students in their undergraduate thesis on different types of particle manipulations including trapping. Prepared Proposal for Research Grant.
- Skills developed: COMSOL Multiphysics, Wolfram Mathematica, OriginPro, Blender

Designation: **Lab Instructor**

Oct 2022 - Jan 2023 (4 months)

- Description: Lab Instructor of 3 one-credit lab of 30 students/section in the Electrical and Computer Engineering Department for 1 trimester.
- Responsibilities: Conducted 3-hours lab classes/section per week. Provided student consultation hours (11 hours/week). Prepared and checked quiz and final exams (Practical & Written). Invigilated the exams. Checked weekly lab reports submitted by the students. Assigned and guided students in projects (Displaying a word on seven segment display with synchronous counter)
- Courses:
 - EEE111L (Analog Electronics)
 - EEE211L & CSE231L (Digital Logic Design)

Designation: Teaching Assistant**2021-2022 (16 months)**

- Description: Teaching Assistant in Electrical and Computer Engineering Department (2 trimesters) and Department of Mathematics and Physics (2 trimesters).
- Responsibilities: Provided weekly consultation hours to students. Checked assignment and quiz papers. Invigilated the exams.
- Courses (4-5 sections/trimester):

○ EEE411 (VLSI)	○ EEE342 (Control Engineering)	○ MAT350 (Engineering Mathematics)
○ EEE413 (Verilog HDL)	○ EEE111 (Analog Electronics)	
○ EEE363 (Electrical Machines)	○ EEE321 (Introduction to Communications Systems)	
	○ MAT116 (Pre-calculus)	
	○ BUS112 (Business Mathematics)	

NOTABLE UNDERGRADUATE PROJECTS

Senior Design Project/Thesis (EEE499 A&B) | North South University**Oct 2021 - Apr 2022**

Project: Implementation of Matter Wave Tractor Beam on COMSOL Multiphysics

- 7-months duration project under the guidance of Dr. Mahdy Rahman Chowdhury in the Electrical and Computer Engineering Department.
- Modeled and verified Matter Wave Tractor Beam on the Schrodinger Equation Interface.
- Presented the project on Capstone Project Showcase at North South University.

Junior Design Project (EEE299) | North South University**Jun 2021 - Sep 2021**

Project: Cloud-based Self-service Ordering System Using Arduino (Kiosk)

- 3-months duration project under the guidance of Dr. Nafisa Noor in the Electrical and Computer Engineering Department.
- Learned to connect different hardware components with Arduino (TFT display, temperature sensors, etc.).
- Skills developed: Arduino ide, Proteus, Overleaf.

TECHNICAL SKILLS

Softwares:

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| <ul style="list-style-type: none">• COMSOL Multiphysics (Schrodinger Equation Interface; Electromagnetic Wave Interface)• Quantum Espresso | <ul style="list-style-type: none">• Multisim• Logisim• Proteus• Arduino ide• Cadence | <ul style="list-style-type: none">• Microwind• Modelsim• Xilinx• OriginPro• Adobe Illustrator and PS• Blender |
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Programming Languages:

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| <ul style="list-style-type: none">• Python (QuTiP, Qiskit)• C++ | <ul style="list-style-type: none">• MATLAB• Mathematica | <ul style="list-style-type: none">• Verilog HDL• Assembly Language |
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Proficiency in Handling Equipments:

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| <ul style="list-style-type: none">• Oscilloscope• DMM | <ul style="list-style-type: none">• Function Generator• Digital Analog Trainer Board | <ul style="list-style-type: none">• Arduino, etc |
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WORKSHOPS AND COURSES OUTSIDE CURRICULUM

Online Courses:

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| <ul style="list-style-type: none">• QBronze (Introduction to Qiskit) by QWorld & Womanium• Optics and Photonics (under Dr. Mahdy Rahman Chowdhury)• Quantum Mechanics (under Dr. Mahdy Rahman Chowdhury) | Jun 2024
2021 (3 months)
2020 (3 months) |
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Workshops:

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| <ul style="list-style-type: none">• Material Characterization (Amalgam, IITM)• Nanoscience and Technology (Amalgam, Techobytes & IITM)• AI Reshaping Material Science (Amalgam, Techobytes & IITM) | 1 Mar 2024
2 Mar 2024
3 Mar 2024 |
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